

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

ELECTRIC STEERING COLUMN LOCK CONTROL MODULE (ESCL)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Electric Steering Column Lock Control Module (ESCL). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Anti-Theft - Passive](#) (419-01B Anti-Theft

- Passive, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B100D-16	Column Lock Authorisation - Circuit voltage below threshold	▪ Battery supply voltage too low ▪ Electric Steering Column Lock circuit short to ground, high resistance, open circuit ▪ Electric steering column lock control module internal failure	▪ Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Refer to the electrical circuit diagrams and check the electric steering column lock control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs. Perform routine - On Demand Self Test (0x0202). If the fault persists, install a new electric steering column lock control module
B100D-29	Column Lock Authorisation - Signal invalid	▪ Battery supply voltage too low ▪ Electric steering column lock control module internal sensor signal invalid	▪ Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, clear the DTCs. Perform routine - On Demand Self Test (0x0202). If the fault persists, install a new electric steering column lock control module
B100D-42	Column Lock Authorisation - General memory failure	▪ Battery supply voltage too low ▪ Electric steering column lock control module internal failure	▪ Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, clear the DTCs. Perform routine - On Demand Self Test (0x0202). Perform a CAN network integrity test. If the fault

			persists, install a new electric steering column lock control module
B100D-51	Column Lock Authorisation - Not programmed	▪ Electric Steering Column Lock not programmed	▪ Using the manufacturer approved diagnostic system, re-configure the electric steering column lock control module as a new module
B100D-62	Column Lock Authorisation - Signal compare failure	▪ Mismatch between electric steering column lock and central junction box security data ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routine - On Demand Self Test (0x0202). If the fault persists, re-configure as a new module ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B100D-64	Column Lock Authorisation - Signal plausibility failure	▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Anti-lock brake system fault ▪ Engine system fault ▪ Body system fault	NOTE: Prior to clearing this DTC, perform the Vehicle Functional Reset routine ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the engine control module for related

			DTCs and refer to the relevant DTC index Using the manufacturer approved diagnostic system, check the central junction box for related DTCs and refer to the relevant DTC index
B100D-72	Column Lock Authorisation - Actuator stuck open	- Battery supply voltage too low - Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely - High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Electric steering column lock control module internal failure	- Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system - Turn the steering wheel to ensure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the manufacturer approved diagnostic system, clear the DTCs and retest - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs. Perform routine - On Demand Self Test (0x0202). If the fault persists, install a new electric steering column lock control module
B100D-73	Column Lock Authorisation - Actuator stuck closed	- Battery supply voltage too low - Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely	- Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system - Turn the steering wheel to ensure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the manufacturer approved

		▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	diagnostic system, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs. Perform routine - On Demand Self Test (0x0202). If the fault persists, install a new electric steering column lock control module
B100D-77	Column Lock Authorisation - Commanded position not reachable	▪ Battery supply voltage too low ▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	▪ Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Turn the steering wheel to ensure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the manufacturer approved diagnostic system, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs. Perform routine - On Demand Self Test (0x0202). If the fault persists, install a new electric steering column lock control module
B100D-92	Column Lock Authorisation - Performance or	▪ Battery supply voltage too low	▪ Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU

	incorrect operation	▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ Electric steering column lock control module internal failure	Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Turn the steering wheel to ensure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the manufacturer approved diagnostic system, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs. Perform routine - On Demand Self Test(0x0202). If the fault persists, install a new electric steering column lock control module
B100D-94	Column Lock Authorisation - Unexpected operation	▪ Battery supply voltage too low ▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	▪ Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Turn the steering wheel to ensure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the manufacturer approved diagnostic system, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved

			diagnostic system, clear the DTCs. Perform routine - On Demand Self Test (0x0202). If the fault persists, install a new electric steering column lock control module
B100D-96	Column Lock Authorisation - Component internal failure	▪ Battery supply voltage too low ▪ Parked position of the road wheels exerting a torque on the steering column, preventing the electric steering column lock mechanism from moving freely ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric steering column lock control module internal failure	NOTE: Prior to clearing this DTC, perform the Vehicle Functional Reset routine ▪ Using the manufacturer approved diagnostic system, check datalogger signal – Main ECU Voltage Supply (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Turn the steering wheel to ensure that the electric steering column lock mechanism is free to move and locking bolt is not restricted. Using the manufacturer approved diagnostic system, clear the DTCs and retest ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs. Perform routine - On Demand Self Test (0x0202). If the fault persists, install a new electric steering column lock control module
U0001-88	High Speed CAN Communication Bus - Bus off	▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit

		resistance	to ground, short circuit to power, open circuit, high resistance
U3002-81	Vehicle Identification Number - Invalid serial data received	- Invalid vehicle identification number - New electric steering column lock installed and VIN not yet programmed	- Using the manufacturer approved diagnostic system, re-configure the electric steering column lock control module with the latest level software - Using the manufacturer approved diagnostic system, perform routine - Learn VIN (0x0404)